

Spam Mail Prediciton

```
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
```

```
data = {
    'Message_ID': [101, 102, 103, 104, 105, 106, 107, 108],
    'Sender_Name': [
        "ABC Bank", "Lottery Team", "John",
        "College", "Reward Center", "Teacher",
        "Gift Shop", "Manager"
    ],
    'Message': [
        "Congratulations! You won a lottery",
        "Free entry in a prize contest",
        "Call me when you are free",
        "Please attend the class tomorrow",
        "You have won a cash reward",
        "Submit your assignment before Monday",
        "Claim your free gift now",
        "Project meeting at 10 AM"
    ],
}
```

```
'Label': [  
    "spam", "spam", "ham", "ham",  
    "spam", "ham", "spam", "ham"  
]  
}
```

```
df = pd.DataFrame(data)
```

```
vectorizer = CountVectorizer()  
X = vectorizer.fit_transform(df['Message'])  
y = df['Label']
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
model = MultinomialNB()  
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)  
print("Model Accuracy:", round(accuracy_score(y_test, y_pred) * 100, 2), "%")
```

```
message_id = input("\nEnter Message ID: ")  
sender_name = input("Enter Sender Name: ")  
message = input("Enter Email Message: ")
```

```
message_vector = vectorizer.transform([message])
```

```
prediction = model.predict(message_vector)
```

```
print("\n----- Prediction Result -----")
```

```
print("Message ID :", message_id)
```

```
print("Sender Name:", sender_name)
```

```
print("Message   :", message)
```

```
print("Prediction :", prediction[0].upper())
```

Output:

Model Accuracy: 100.0 %

Enter Message ID: 201

Enter Sender Name: Lottery Team

Enter Email Message: Congratulations! You have won a free gift.

----- Prediction Result -----

Message ID : 201

Sender Name: Lottery Team

Message : Congratulations! You have won a free gift.

Prediction : SPAM